Math 421

Monday, October 6

Announcements

- Homework 5 due Friday
  - Any [redacted] theorems proven or stated in class, readings, homework assignments, and other course related materials, can be cited and used without proof.
- Grace's Drop-In Hours
  - Wednesday: 12-1 VV 311
  - Thursday: 9-11 VV B205
- MLC
  - Proof Table: M-Th 3-7:30 VV B227
  - Course Assistant: Th 4-7 Table 4
- Exam 1 – 10/15 5:45 – 7:15 pm
  - If you have a conflict, email me (if you haven't already) and **fill out makeup exam request form** (linked from Exam Information page on Canvas) ASAP!

Chapter 2: Sequences

Section 9: Limit Theorems for Sequences

Bounded

Definition. A sequence (s_n) is said to be **bounded** if the set $\{s_n : n \in \mathbb{N}\}$ is a bounded set.

$\hookrightarrow$ there exists $M \in \mathbb{R}$, such that $\{s_n : n \in \mathbb{N}\} \subseteq [-M, M]$ or $|s_n| \leq M$

Activity –

1. Give an example of a bounded and an unbounded sequence.

   $(\sin(n))$ $\quad \{\frac{1}{n} : n \in \mathbb{N}\} \subseteq [-1, 1]$ $\quad (n)$

2. Does every bounded sequence converge? Does every unbounded sequence diverge?

   No. $(\sin(\frac{n\pi}{4}))$ $\quad$ Yes.

Theorem 9.1

$\lim s_n = s$. for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ s.t. $n > N$, $|s_n - s| < \varepsilon$.

must find

there exists $\boxed{M \in \mathbb{R}}$ such that $|s_n| \leq M$ for all n.

Theorem. *Convergent sequences are bounded.* (s_n)

Discussion.

Recall: $|s_n| - |s| \leq |s_n - s| < 1$ choice. could be any $\oplus$ #.

$\Rightarrow |s_n| < 1 + |s|$ for $n > N$

~~$M = 1 + |s|$~~ ?

$M = \max\{1 + |s|, |s_1|, |s_2|, \dots |s_N|\}$

$|s_n| = |s_n - s + s| \leq |s_n - s| + |s|$

Proof.

Let (s_n) be a convergent sequence. So $\lim s_n = s$ for some $s \in \mathbb{R}$. Since $\lim s_n = s$, there exists $N \in \mathbb{N}$ such that for all $n > N$, $|s_n - s| < 1$. Then $|s_n| < |s| + 1$. Let $M = \max\{1 + |s|, |s_1|, |s_2|, \dots, |s_N|\}$. Then for all n, $|s_n| \leq M$, so (s_n) is bounded.

Algebraic Limit Laws

Theorems 9.2-9.6: Let $\lim s_n = s$ and $\lim t_n = t$. Then,

i. $\lim ks_n = ks$ for all $k \in \mathbb{R}$. ← discussed in class.

ii. $\lim s_n + t_n = s + t$. ← proven in text.

Today: false proofs.

iii. $\lim s_n t_n = st$.

iv. if $s_n \neq 0$, for all n, and if $s \neq 0$, then $\lim \frac{1}{s_n} = \frac{1}{s}$

v. if $s_n \neq 0$, for all n, and if $s \neq 0$, then $\lim \frac{t_n}{s_n} = \frac{t}{s}$

FALSE Proof of ii. $\lim s_n + t_n = s + t$.

Activity: Critique the proof below.

Proof. Fix $\epsilon > 0$. Since $\lim s_n = s$, there exists $\boxed{N_1 > 0}$ such that if

might not be equal

$n > N_1$, then $|s_n - s| < \frac{\epsilon}{2}$. Since $\lim t_n = t$, there exists $\boxed{N_2 > 0}$ such that if $n > N_2$, then $|t_n - t| < \frac{\epsilon}{2}$. Assume $\cancel{n > N}$ $n > \max\{N_1, N_2\}$. Then

$$|(s_n + t_n) - (s + t)| = |(s_n - s) + (t_n - t)| \leq |s_n - s| + |t_n - t|$$

$$< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

Since $\epsilon > 0$ was arbitrary, $\lim s_n + t_n = s + t$.

FALSE Proof of iii. $\lim s_n t_n = st$.

Activity: Critique the proof below.

Proof. Fix $\epsilon > 0$. Since $\lim s_n = s$, there exists $N_1 > 0$ such that if $n > N_1$, then $|s_n - s| < \frac{\epsilon}{2(|t|+1)}$. Since $\lim t_n = t$, there exists $N_2 > 0$ such that if $n > N_2$, then $|t_n - t| < \frac{\epsilon}{\boxed{2\cancel{|s_n|}}\ M}$. Let $N = \max\{N_1, N_2\}$.

must be fixed, cannot include n.

Assume $n > N$. Then

$$|s_n t_n - st| = |s_n t_n - s_n t + s_n t - st| \le |s_n t_n - s_n t| + |s_n t - st|$$

$$\le |s_n||t_n - t| + |t||s_n - s| < \overset{M}{\cancel{|s_n|}}\frac{\epsilon}{2\underset{M}{\cancel{|s_n|}}} + (|t|+1)\frac{\epsilon}{2(|t|+1)} = \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

Since $\epsilon > 0$ was arbitrary, $\lim s_n t_n = st$.